

Affect and favorable work outcomes: two longitudinal tests of the happy– productive worker thesis

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Summary

This research examined relationships between alternative measures of affect and supervisory performance ratings. The first study showed that dispositional rather than state affect significantly predicted supervisory ratings of performance over time. Since the measures of affect differed on both content and temporal dimensions, a follow-up study was conducted to explicate the results. The second study found that a pleasantness-based measure of dispositional affect (Berkman, 1971a) again predicted rated performance over time, but activation-based measures of both dispositional and state affect (using PANAS scales) were not predictive of supervisory evaluations of performance. The implications of these findings in terms of research on affect and the longstanding pursuit of the happy–productive worker are discussed. Copyright © 1999 John Wiley & Sons, Ltd.

Introduction

[Management] tries to make Southwest a fun place to work, knowing that happy employees are also productive ones. (*San Francisco Chronicle*, June 10, 1993, page D3)

Most researchers have relegated the notion of the happy–productive worker to the folklore of management—as an unsubstantiated claim of practitioners and the popular press (Staw, 1986). Because decades of research have failed to show a close link between job satisfaction and performance (see Brayfield and Crockett, 1955; Vroom, 1964; Locke, 1976; Iaffaldano and Muchinsky, 1985), for most part of the past, the field of organizational behavior has ceased investigating whether happier workers are also more productive.

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Recently, however, some interesting research on the broader construct of psychological affect has started to accumulate. As reviewed by Isen and Baron (1991) and George and Brief (1992), individuals experiencing positive affective states have been found to be more helpful, more creative, better negotiators, and more persistent on uncertain tasks. Though affect has rarely been examined as a predictor of more traditional measures of favorable work outcomes (see George and Bettenhausen (1990) for an exception), the implication of this research is clear. It is at least plausible that affective states and work behavior could be linked, thereby reopening the question of whether happy workers are also more productive.

Why Affect and Performance May be Linked

George and Brief (1996) recently used long-established models of motivation to hypothesize how positive affect could facilitate employee achievement or performance. They argued, using expectancy theory (e.g. Vroom, 1964), that positive affect might enhance both the expectancy that one's effort leads to performance and the belief that performance leads to positive outcomes. The basis of these predictions comes from research showing that positive affect can lead to greater self-efficacy (Forgas, Bower and Moylan, 1990) and optimistic biases in the estimation of future events (Seligman, 1991; Wright and Bower, 1992). An analogous hypothesis could also be formulated using a goal-setting model of motivation (Locke and Latham, 1990). Here one might posit that optimistic tendencies will lead individuals to set more difficult goals for themselves and/or accept more challenging goals provided by others, thus contributing to higher performance. In a similar fashion, one might also use an attributional model of motivation (Weiner, 1985) to hypothesize that affect has a facilitative influence on task persistence (George and Brief, 1996). Because people who are high in positive affect or optimism tend to interpret failure as a temporary setback caused by external circumstances (Forgas, 1992), they are more likely to persevere following adverse feedback than those with negative affective or pessimistic tendencies (Burke, Brief and George, 1993).

The influence of affect may also go beyond the motivational aspects of task performance. Since positive affect has been shown to be a determinant of helping behavior and interpersonal cooperation, it may be predictive of a number of extra-role or citizenship behaviors within the organization (George and Brief, 1992; Organ, 1988). Likewise, having a positive mood or disposition can lead to success via the help one gets from others in an organization. Because people are more attracted to positive than negative people, those with positive affect are more likely to be offered social support (Caplan, Cobb, French, Harrison and Pinneau, 1980; Ganster, Fusilier and Mayes, 1986), selected as part of a political or supervisory in-group (Graen, 1976), picked for valued positions (Imada and Hakel, 1977; McGovern and Tinsley, 1978; Rasmussen, 1984), and evaluated positively by others in the organization (Cardy and Dobbins, 1986; Krzystofiak, Cardy and Newman, 1988). Thus, individuals displaying positive affect may be more successful, not only because of their work effort or output, but because they are more adept at the interpersonal aspects of organizational life.

In this research we will not sort out the various paths or mechanisms by which individuals become successful. We test only whether affect is associated with performance *as it is perceived or operationalized by the organizational members*. This is a necessary first step for research on affect and performance—one that was never fulfilled by earlier research using job satisfaction as the predictor variable. Thus, the present studies should be interpreted as an examination of the *plausibility* (rather than proof) of the happy-productive worker thesis. If there is evidence for an

affect–performance relationship, further investigation will be necessary on the means by which happy workers actually become successful in organizational life.

Forms of Affect

Although this research does not delve into the meaning or facets of organizational performance, it does consider some alternative forms of affect. Affect is frequently divided into positive and negative factors, with separate hypotheses for each of these dimensions (Burke *et al.*, 1993; Watson and Tellegen, 1985). The division of affect into two unipolar dimensions is still methodologically controversial (see Green, Goldman and Salovey, 1993), however. Larsen and Diener (1992) have noted that questions phrased positively often form a factor separable from those phrased negatively. They have also noted that what Watson and others have labeled as unipolar dimensions of positive and negative affect are actually bipolar scales, and that these factors generally tap activation (or stress) rather than hedonic (happiness–sadness) aspects of affect. Thus, it is unclear whether tests of the happy–productive worker thesis will be stronger using separate, unipolar measures of positive and negative affect (e.g. Watson, 1988) or more conventional bipolar scales. We will therefore use both forms of affect in this research.

A second aspect of affect that merits attention is its temporal nature. To argue that ‘a happy worker is a productive one’ says little about whether happiness is a state or trait variable. Clearly, most of the psychological literature showing that affect can influence outcomes such as helping, cooperativeness, or creativity has relied on the manipulation of short-term affective states, usually within laboratory settings (George, 1992; Isen and Baron, 1991). Yet, many of the organizationally grounded studies that are supportive of the happy–productive worker thesis have used dispositional measures of affect. We will briefly review these studies and describe why dispositional rather than state affect may be the stronger operationalization of the happy–productive worker thesis.

Prior Research on the Affect–Performance Link

Several studies have shown that poor mental health is related to alcoholism, drug consumption, frequent job change, hypertension, loss of self-esteem, person–environment (mis)fit and role stress (Cooper and Marshall, 1976; Furnham and Schaeffer, 1984; Isaksson, 1990; Kelloway and Barling, 1991; Martin, 1984). Since these variables have, in turn, been related to declines in work outcomes (Ivancevich and Matteson, 1980; Quick and Quick, 1984), it is possible that dispositional affect and measures of employee performance are at least indirectly linked.

More direct evidence that dispositional affect can predict indicators of employee performance or success comes from four studies. Seligman and Schulman (1986) divided a sample of 103 new life insurance agents on measures of dispositional optimism, and found that the optimists remained on their jobs at twice the rate of pessimists and sold more insurance than pessimists. Staw, Sutton and Pelled (1994) reanalyzed a longitudinal data set to examine whether a measure of depression and other affect items could predict changes in work outcomes. Staw *et al.* found that their composite measure of affect successfully predicted changes in salary, performance evaluation, and social support. Staw and Barsade (1993) used Tellegen’s (1982) measure of

well-being as well as self and observer ratings of affect to predict performance on in-basket exercises. Staw and Barsade found that dispositional affect was a significant predictor of both decisional and interpersonal aspects of performance. Finally, Wright, Bonett and Sweeney (1993) found moderate support for a positive relationship between a dispositional measure of well-being and work-related performance.

The studies supportive of the happy-productive worker thesis appear to involve dispositional rather than state affect (Cropanzano, James and Konovsky, 1993). One reason for this disparity may be methodological. When affective state is manipulated experimentally (e.g. by receiving a free gift or seeing an upbeat film), the predicted behavior (or dependent variable) must be measured shortly thereafter, since mood alterations are usually short-lived. In contrast, performance in most work settings is not so easily delimited in time. Except in the case of sales and production work (George and Bettenhausen, 1990), it is difficult to monitor the work output of employees so that mood and performance can be precisely linked. Typically, performance ratings as well as pay and promotion decisions are infrequent events designed to reflect months (or years) of previous employment. Therefore, one would expect measures of dispositional affect (designed to assess affect over long stretches of time) to provide a closer fit with most individual-based performance measures than assessments of mood or affective state.

There may also be theoretical reasons for expecting dispositional affect to be the better predictor of individual measures of employee performance or success. Dispositional affect may have a greater influence than mood because it is more stable and enduring over time, allowing continual as opposed to fleeting effects on cognition and behavior. Because an affective state is not long-lasting, it may be replaced by another, perhaps contrary, state before stable cognitions (e.g. self-efficacy) can be formulated. In addition, since there is likely delay between holding certain cognitions (e.g. expectancy of success) and opportunity to enact an associated behavior (e.g. volunteering for a difficult assignment), any alteration or fading of an affective state would decrease its likelihood of influence. Thus, there is theoretical as well as methodological justification for expecting a stronger linkage of dispositional rather than state affect with measures of individual performance.

Research Hypotheses

We tested two alternative versions of the happy-productive worker thesis, linking both state affect (or mood) as well as trait (or dispositional) affect with performance.

Thus,

Hypothesis 1: The individual's affective state will be directionally associated with supervisory performance evaluations at a particular period of time.

Hypothesis 2: The individual's affective disposition will be directionally associated with supervisory performance evaluations over multiple periods of time.

Overview of the Research

The research consisted of two longitudinal field studies. In Study 1, both dispositional and state measures of affect were used to predict supervisory evaluations of performance. Because of the

difficulties in tying state affect to performance ratings at a specific point in time, we considered the mood–performance relationship as an exploratory or secondary test of the happy–productive worker thesis. In Study 2, we improved upon the calibration of measurement for testing the state affect hypothesis and compared two dispositional affect measures in their ability to predict performance.

Study 1

Method

The measurement design of Study 1 is shown in Figure 1. Data were collected at four time periods: T1 (first year of the study), T2 (third year of the study), T3 (early in the fourth year of the study), and T4 (later in the fourth year of the study). Measures of affective disposition were collected at times T1 and T2. Measures of positive and negative moods were collected at T2 and T3. Supervisory evaluations of composite and global performance were available for T2, T3, and T4.

Subjects

All individuals employed within a public sector, social welfare department ($N = 81$) of a major metropolitan city in California were asked to participate in Study 1. These departmental employees were classified as professional (e.g. 4-year college degree requirement) and all performed their job duties under the same general job description. The sample for this study included 53 social welfare staff, representing a response rate of 65.4 per cent. The subjects were representative of the department's population in that they were primarily male (71.7 per cent), with a mean age of 45.3 years and a mean job tenure of 16.7 years.

In longitudinal research, where subjects are difficult to retain over repeated measurements, it is important to determine the minimum number of subjects necessary for the detection of a meaningful effect size. Guilford (1956, p. 145) noted that an effect size of 0.6 is indicative of a substantial relationship. Since prior research by Wright *et al.* (1993) found simple correlations between dispositional affect and performance ratings in the 0.4 range, it was not unreasonable to expect a multiple correlation to approximate 0.6. Therefore, using a standard sample size determination procedure (Cohen, 1988), a sample size of at least 30 would be required to detect a multiple correlation of 0.6 with power equal to 0.75 and alpha equal to 0.01. Thus, the obtained sample sizes used in the two regression analyses (e.g. 50 and 51, respectively), though small in comparison to many cross-sectional studies, can be considered adequate for this longitudinal research.

Variables	T1	T2	T3	T4
Affective disposition	X	X		
Rated performance		X	X	X
Positive mood		X	X	
Negative mood		X	X	

Figure 1. Measurements design for study 1. Data were collected at four time periods: T1 (initial), T2 (year 3), T3 (year 4) and T4 (year 4 1/2). The sample includes all subjects for whom measures of affective state, affective disposition and prior rated performance were available as predictors of subsequent rated performance

Measures

Affective state To assess state affect or mood of the individual we used the PANAS scale developed by Watson, Clark and Tellegen (1988b). The PANAS scale is designed to measure both positive affect (PA) and negative affect (NA). PA is measured by descriptors such as 'active, alert, enthusiastic, inspired, and interested'. NA is assessed by descriptors such as 'afraid, hostile, irritable, jittery, and upset'. Subjects indicated the extent to which they experienced each mood state *that day* (e.g. today) on a 5-point scale ranging from 'very slightly or not at all' to 'extremely'. Extensive research has demonstrated the reliability and validity of the instrument across a wide range of subject samples (Watson, 1988; Watson *et al.*, 1988b). In Study 1, Cronbach's alpha for positive mood was 0.91 at T2 and 0.86 at T3; for negative mood alpha was 0.93 at T2 and 0.90 at T3.

Affective disposition As a measure of affective disposition this study utilized the 8-item Index of Psychological Well-Being developed by Berkman (1971a,b). Like many other well-being measures (Diener, 1984), this scale was designed to assess people's mental health on a single affective index. The Berkman scale uses many of the same items as Bradburn and Caplovitz' (1965) classic measure of affect, but with a more general or open-ended time horizon. Subjects were asked how often they felt: 'very lonely or remote from other people', 'depressed or very unhappy', 'bored', 'so restless you couldn't sit long in a chair', 'vaguely uneasy about something without knowing why', 'particularly excited or interested in something', 'pleased about having accomplished something' and 'on top of the world'. The reader is referred to Berkman (1971a,b) and Wright and Bonett (1992) for a further description of the scoring and prior validation using a probability sample of 6928 adults from Alameda County, California. For the current sample, Cronbach's alphas for the dispositional affect scale were established at 0.73 (T1) and 0.70 (T2). The test-retest reliability (T1-T2) coefficient was 0.74.¹

Supervisor evaluations Two measures of employee performance or success were available in this research at T2, T3 and T4. The first measure was a 1-item, global rating of performance. The top-ranking administrative officer was asked 'Overall, how would you rate this employee's performance *at this time*?' A 5-point response scale was provided with ratings ranging from 'poor' to 'excellent'.

The second supervisory performance evaluation measure was a composite scale of performance developed by Wright and his associates (Wright and Bonett, 1993; Wright *et al.*, 1993). Management personnel from the present organization confirmed four dimensions relevant for assessing the achievement or performance of the social welfare personnel sampled: work facilitation, goal emphasis, support, and team building. Because all employees ultimately reported to the same top-ranking administrative officer, this departmental manager rated each employee's achievement for the past year at T2 and T3 and for the past 6 months at T4.

¹ The mean of dispositional affect at times 1 and 2 was 3.5 for Study 1. For Study 2, the means at times 1 and 2 were 3.6 and 3.4, respectively. The standard deviations of dispositional affect at times 1 and 2 were 1.3 and 1.5 for Study 1. For Study 2, the standard deviations at times 1 and 2 were 1.3 and 1.4, respectively. Tests for the equality of correlated means found that the two means are not significantly different for either Study 1 ($t = 1.28$; $df = 35$; $p = 0.2104$) or Study 2 ($t = 1.22$; $df = 61$; $p = 0.2278$). Furthermore, tests for the equality of correlated variances did not reveal any differences for either Study 1 ($t = -1.356$; $df = 32$; $p = 0.1847$) or Study 2 ($t = -0.623$; $df = 60$; $p = 0.5355$). These findings indicate that the two dispositional affect scores satisfy the definition of a strictly parallel measurement model (Kristof, 1963), and confirm that the best estimate of dispositional affect is the average of the two dispositional affect scores (Wright *et al.*, 1993).

The composite measure asked for an assessment of the individual (using 5-point scales) on the following work dimensions: 'To what extent does this employee offer useful ideas for solving technical/administrative problems?' (work facilitation); 'To what extent does this employee develop and maintain high performance goals?' (goal emphasis); 'To what extent does this employee pay attention to what you are saying when talking to them?' (support); 'To what extent does this employee work harmoniously with his/her co-worker(s) as a team to achieve group/organizational goals and purposes?' (team building). Ratings were summed to form a composite measure of achievement or performance for each subject (coefficient alphas = 0.76 for T2; 0.81 for T3; 0.81 for T4).

Because of its more specific time referent, the global measure of rated performance is the most relevant for testing the relationship between state affect and rated performance. Of course, even with specific time referents it is possible for current supervisory ratings to be colored by past performance ratings. Thus, the examination of hypothesis 1 should be viewed as a more limited and exploratory inquiry rather than a definitive test of the influence of state affect on performance ratings. Because dispositional affect is more stable over time, temporal discrepancies are less likely to weaken the tests of hypothesis 2, namely that dispositional affect is predictive of rated performance over time.

Results

The means, standard deviations, and intercorrelations of each of the measures are reported in Table 1.

Affective states and supervisory evaluations

Table 1 shows the correlations of both positive and negative mood to the performance ratings. Because mood is considered a transient state, the most relevant tests of hypothesis 1 are the relationships of mood to rated performance during the time periods in which both variables were measured contemporaneously. This resulted in eight correlations examining the relationships between both positive and negative affect with the global and composite performance ratings at times 2 and 3. However, as already noted, primary emphasis should be placed upon the four relationships between affect and global ratings of performance, since these performance ratings contained a specific temporal referent (*at this time*). Out of all the correlations with either global or composite performance ratings, only one (between NA and global ratings of performance at T2) was statistically significant.

Affective disposition and supervisory evaluations

Table 1 also reports the correlations between dispositional affect, at T1 and T2, and composite supervisory performance ratings at T2, T3 and T4. Significant positive relationships were found between dispositional affect at T1 and subsequent composite supervisory performance ratings at T2, T3, and T4. Similarly, significant positive relationships between dispositional affect at T2 and composite performance ratings at T2, T3, and T4 were obtained. Out of a total of 12 correlations relating affective disposition to either the composite or global measure of rated performance, 11 were statistically significant in the predicted direction, with the remaining correlation reaching marginal significance ($p < 0.06$).

Table 1. Means, standard deviations, and Pearson correlations for Study 1*

Variables	<i>N</i>	<i>M</i>	<i>S.D.</i>	1	2	3	4	5	6	7	8	9	10	11	12
1. Time 1 affective disposition	44	3.5	1.3	—	0.74†	0.45‡	0.39‡	0.47§	0.36‡	0.25	0.37‡	0.49‡	0.49‡	−0.31	−0.39
2. Time 2 affective disposition	45	3.5	1.5		—	0.33	0.35	0.48§	0.32	0.33	0.37‡	0.41‡	0.42‡	−0.46‡	−0.44‡
3. Time 2 composite rated perf.	50	3.6	0.6			—	0.67†	0.60†	0.62†	0.48§	0.58†	0.05	0.23	−0.11	−0.16
4. Time 3 composite rated perf.	51	3.4	0.6				—	0.64†	0.63†	0.69†	0.67†	−0.13	0.09	−0.31	−0.13
5. Time 4 composite rated perf.	53	3.5	0.6					—	0.41‡	0.47§	0.65†	0.08	0.16	−0.23	−0.14
6. Time 2 global rated perf.	52	3.9	0.6						—	0.66†	0.67†	−0.05	0.13	−0.30	−0.35
7. Time 3 global rated perf.	51	4.0	0.7							—	0.77†	0.08	0.13	−0.39‡	−0.12
8. Time 4 global rated perf.	53	4.1	0.8								—	0.03	0.21	−0.40‡	−0.12
9. Time 2 positive mood	41	3.1	0.8									—	0.58§	−0.24	−0.11
10. Time 3 positive mood	43	3.2	0.7										—	0.28	−0.14
11. Time 2 negative mood	41	1.9	0.9											—	0.64†
12. Time 3 negative mood	43	1.6	0.7												—

* One-tailed tests.

† $p < 0.0001$; ‡ $p < 0.01$; § $p < 0.001$; || $p < 0.05$.

Table 2 shows an aggregation of dispositional affect and its relationship with supervisory performance ratings over time. As one might expect (e.g. Epstein, 1979), the aggregated measures of dispositional affect were somewhat stronger predictors of rated performance than were dispositional measures from a single time period. Aggregating performance rating measures over time also strengthened the relationship between dispositional affect and supervisory ratings of performance.

Table 3 shows the correlations between average dispositional affect and the four items making up the composite measure of rated performance. Significant positive relationships between dispositional affect and support and goal emphasis were found at T2, T3, and T4. There were also

Table 2. Correlations of average affective disposition (Berkman scale) and rated performance for studies 1 and 2

Study 1		
Average affective disposition (1 & 2) and composite rated perf. at T2:	0.38*	N = 50
Average affective disposition (1 & 2) and global rated perf. at T2:	0.33*	N = 52
Average affective disposition (1 & 2) and composite rated perf. at T3:	0.40*	N = 51
Average affective disposition (1 & 2) and global rated perf. at T3:	0.29†	N = 51
Average affective disposition (1 & 2) and composite rated perf. at T4:	0.51‡	N = 53
Average affective disposition (1 & 2) and global rated perf. at T4:	0.38*	N = 53
Average affective disposition (1 & 2) and composite rated perf. at (T2 & T3):	0.42*	N = 51
Average affective disposition (1 & 2) and composite rated perf. at (T3 & T4):	0.52‡	N = 53
Study 2		
Average affective disposition (1 & 2) and composite rated perf. at T2:	0.48§	N = 56

* $p < 0.01$; † $p < 0.05$; ‡ $p < 0.001$; § $p < 0.0001$ —one-tailed tests.

Table 3. Correlations of average affective disposition (Berkman scale) and each rated performance dimension for Studies 1 and 2

Study 1	
Average affective disposition (1 & 2) and support at T2:	0.35*
Average affective disposition (1 & 2) and support at T3:	0.30†
Average affective disposition (1 & 2) and support at T4:	0.28†
Average affective disposition (1 & 2) and goal emphasis at T2:	0.41*
Average affective disposition (1 & 2) and goal emphasis at T3:	0.28†
Average affective disposition (1 & 2) and goal emphasis at T4:	0.47‡
Average affective disposition (1 & 2) and team building at T2:	0.17
Average affective disposition (1 & 2) and team building at T3:	0.32†
Average affective disposition (1 & 2) and team building at T4:	0.46‡
Average affective disposition (1 & 2) and work facilitation at T2:	0.24†
Average affective disposition (1 & 2) and work facilitation at T3:	0.36*
Average affective disposition (1 & 2) and work facilitation at T4:	0.42‡
N = 50 for T2; 51 for T3; 53 for T4.	
Study 2	
Average affective disposition (1 & 2) and support at T2:	0.30†
Average affective disposition (1 & 2) and goal emphasis at T2:	0.46‡
Average affective disposition (1 & 2) and team building at T2:	0.41‡
Average affective disposition (1 & 2) and work facilitation at T2:	0.44‡
N = 56	

* $p < 0.01$; † $p < 0.05$; ‡ $p < 0.001$ —one-tailed tests.

significant relationships between dispositional affect and team building at T3 and T4. Finally, significant relationships between affect and work facilitation at T2, T3, and T4 were obtained.

Though the results in Tables 2 and 3 are longitudinal, they cannot rule out the possibility that performance ratings may have influenced affective disposition over time. In order to help control for the prior and concurrent effects of rated performance on affect, two multiple regression analyses were performed. In each of these analyses composite performance ratings were regressed on both prior measures of rated performance and dispositional affect. The first regression equation considered average rated performance at T3 and T4 as the dependent variable, with prior rated performance at T2 and average dispositional affect at T1 and T2 as the independent variables. The second regression equation considered subsequent rated performance at T4 as the dependent variable, with average prior rated performance at T2 and T3 and average dispositional affect at T1 and T2 as the independent variables.

Since regression coefficients are functions of semi-partial correlations, in the above analyses the effect of prior rated performance will be partialled out of the dispositional affect measure and the effect of dispositional affect will be partialled out of the prior rated performance measure. If dispositional affect is a significant predictor of future ratings of performance, over and above the effects of prior ratings of performance, then one can have greater confidence in its causal influence over time (Pedhazur, 1982).

For the first regression analysis, prior rated performance at T2 and average dispositional affect at T1 and T2 were found to predict average subsequent rated performance at T3 and T4 ($F = 30.08$; $df = 2, 47$; $p = 0.0001$; adjusted $R^2 = 0.54$). One-tailed tests of each predictor variable were significant: for prior rated performance, $t = 5.67$; $df = 47$; $p = 0.0001$; for average dispositional affect, $t = 2.76$; $df = 47$; $p = 0.0041$. As shown in Table 4, these results illustrated that dispositional affect could significantly predict supervisory performance ratings over time.

Table 4. Study 1 results of longitudinal multiple regression: predicting average rated performance (times 3 and 4) from rated performance at time 2 and average affective disposition (times 1 and 2)

Independent variable	Beta	<i>S.E.</i>	<i>t</i>	<i>p</i> Value (one-tailed)
Rated performance (time 2)	0.59*	0.10	5.67	<0.0001
Average affective disposition (times 1 & 2)	0.49†	0.18	2.76	<0.01

$R = 0.75$; $R^2 = 0.56$; adjusted $R^2 = 0.54$.

* $p < 0.0001$; † $p < 0.01$.

$N = 50$.

For the second regression analysis, average prior rated performance at T2 and T3 and average dispositional affect at T1 and T2 were found to predict subsequent rated performance at T4 ($F = 25.12$; $df = 2, 48$; $p = 0.0001$; adjusted $R^2 = 0.49$). One-tailed tests of each predictor variable, average prior rated performance ($t = 5.19$; $df = 48$; $p = 0.0001$) and average dispositional affect ($t = 2.21$; $df = 48$; $p = 0.0158$), were significant. These results, shown in Table 5, also confirmed that dispositional affect significantly predicted supervisory performance ratings beyond that accounted for by the prior supervisory performance ratings.

Discussion

The results of Study 1 provided mixed support for a revised formulation of the happy-productive worker thesis.

Table 5. Study 1 results of longitudinal multiple regression: predicting rated performance (times 4) from average rated performance (times 2 and 3) and average affective disposition (times 1 and 2)

Independent variable	Beta	S.E.	<i>t</i>	<i>p</i> Value (one-tailed)
Average rated performance (times 2 & 3)	0.72*	0.14	5.19	<0.0001
Average affective disposition (times 1 & 2)	0.49†	0.22	2.21	<0.02

$R = 0.71$; $R^2 = 0.51$; adjusted $R^2 = 0.49$.

* $p < 0.0001$; † $p < 0.05$.

$N = 51$.

The first hypothesis, that state affect would covary significantly with rated performance, was not upheld. Positive mood was more positively associated with rated performance than negative mood, but only one of the correlations reached statistical significance. In the face of such a lack of results, it is certainly possible to argue that the data might have been stronger had there been better calibration in the measurement of mood and rated performance. Alternatively, the dispositional measure of affect was a more successful predictor of rated performance. Using both zero-order correlations and longitudinal regression analyses, the dispositional affect measure significantly predicted rated performance, thus supporting hypothesis 2.

It is tempting to conclude from the pattern of results that dispositional affect is a stronger predictor of rated performance than state affect. Yet, because the measures of affect differed in both *content* and *time period*, it is difficult and inappropriate to draw a definitive conclusion. It is possible, for example, that two entirely different affective constructs were tapped in this research. As some scholars have noted (Larsen and Diener, 1992), the PANAS scale used to measure mood contains a large portion of items denoting activation rather than hedonic tone. For example, among the 20 items on the PANAS scale are the following arousal terms: 'active', 'attentive', 'determined', 'excited', 'scared', 'alert', 'nervous', 'jittery' and 'afraid'. Conspicuously missing from the PANAS scale are items such as 'happy', 'contented', 'pleased', 'unhappy', or 'sad'—descriptors that many researchers and the general public associate with the notion of affect. In contrast, Berkman's (1971a) well-being measure contains a greater proportion of items denoting hedonic tone or mental health (e.g. 'depressed or very unhappy', 'very lonely or remote from other people', 'pleased about having accomplished something', and 'on top of the world'). Thus, it is possible that the Berkman measure was a stronger predictor of rated performance than the PANAS scale simply because it more closely taps the pleasantness or hedonic side of affect.

Study 2

Study 2 was designed to test mood versus dispositional predictors of rated performance using measures with the same affective descriptors, varying only on the time dimension. Because the PANAS measure has been previously validated as both a trait and state measure of affect (Watson *et al.*, 1988b), it was used for these mood versus dispositional comparisons. Additional care was taken to provide a good temporal match between mood and performance measures by assessing both mood and rated performance for the 'past week'.

In an effort to replicate the results from Study 1, we again tested whether Berkman's measure of well-being was a significant predictor of performance over time. We intended to compare this

pleasantness-based indicator with the more activation-based PANAS scale in predicting employee performance ratings.² Thus, the specific hypotheses tested in Study 2 were:

Hypothesis 3: The individual's affective mood state (positive or negative) will be a stronger predictor of supervisory performance evaluations than his or her affective disposition.

Hypothesis 4: The individual's affective disposition (positive or negative) will be a stronger predictor of supervisory performance ratings than his or her mood state.

Hypothesis 5: Dispositional affect measured by a pleasantness-based scale will be a stronger predictor of supervisory performance evaluations than dispositional affect measured by a more activation-based scale.

Method

Subjects

Social services staff personnel from a county agency located in northern California ($N = 106$) were asked to participate in Study 2 through a direct contact procedure. The actual sample for Study 2 included 78 employees, representing a response rate of 74 per cent. As with Study 1, respondents were well-educated, over half having a Master's degree or equivalent, with the remaining employees having a Bachelor's degree (both the agencies sampled from Studies 1 and 2 require a minimum of a Bachelor's degree). Ages ranged from 24 to 60 years, the mean age was 39.7 years ($S.D. = 7.5$ years). Job tenure ranged from 1 to 26 years; the mean job tenure was 10.3 years ($S.D. = 5.8$ years). All subjects performed their job duties under the same general job description.

Data for Study 2 were collected at two time periods—T1 (initial) and T2 (approximately 1 year later). The largely pleasantness-based measure of affective disposition (Berkman scale) was administered at T1 and T2 and data were available from 72 subjects. The more activation-based measure of affective disposition (PANAS scale) was separately administered at T2 and data were available from 53 subjects; the activation-based measure of state affect (PANAS scale) was separately administered at T1 and data were available for 43 subjects. Measures of composite performance evaluations were available at T1 for 62 subjects and at T2 for 64 subjects; a global measure of rated performance was available at T1 for 64 subjects. The sample includes all subjects for whom measures of pleasantness or activation-based affect were available as predictors of rated performance.

Measures

Affective state As in Study 1, to assess state affect or mood of the individual, we used the PANAS scale (Watson *et al.*, 1988b). In Study 2, subjects indicated the extent to which they experienced each mood state during the *past week* on a 5-point scale ranging from 'very slightly

² We use the terms 'pleasantness-based' and 'activation-based' in describing the Berkman and PANAS measures, although neither scale reflects only the hedonic or arousal dimensions of an emotional circumplex (e.g. Russell, 1980). We use these terms primarily to highlight what we believe are major content differences between the Berkman and PANAS scales.

or not at all' to 'extremely'. For this study, Cronbach's alpha for positive mood was 0.86; for negative mood alpha was 0.89.

Affective disposition Study 2 utilized two measures of affective disposition. For the pleasantness-based measure of affect we again used Berkman's (1971a,b) Index of Psychological Well-Being. Cronbach's alphas for the Berkman scale were established at 0.67 (T1) and 0.72 (T2). The test-retest reliability (T1–T2) coefficient was 0.77. For the activation-based dispositional measure we used the trait PANAS scale developed by Watson *et al.* (1988b). As noted earlier, PANAS is scored in terms of both positive (PA) and negative affect (NA). PA and NA are measured with the same descriptors for trait and state affect. However, for the trait indicator subjects indicated the extent to which they experienced each positive and negative descriptor of affect *in general* on a 5-point scale ranging from 'very slightly or not at all' to 'extremely'. In Study 2, Cronbach's alpha for positive affectivity was 0.91; alpha for negative affectivity was 0.89.

Supervisor evaluations Two measures of employee performance or success were available for Study 2. The first measure was a 1-item, global rating of performance. Two top-ranking administrative officers were asked, 'Overall, how would you rate this employee's performance *over the past week?*' A 5-point response scale was provided with ratings ranging from 'poor' to 'excellent'. This supervisory rating of global performance was available at T1.

As in Study 1, the second performance measure used the evaluation procedure validated by Wright and Bonett (1993). Management personnel confirmed the four dimensions (work facilitation, goal emphasis, support, and team building) as appropriate and relevant for assessing employee achievement or performance. Two top administrative officers provided these employee performance ratings at T1 and T2. Following Wright and Bonett (1993), the four items, each examining a separate dimension, were summed to form a composite measure of performance (T1 coefficient alpha = 0.79; T2 coefficient alpha = 0.86).

It was possible in Study 2 to assess the degree to which both raters differed in their evaluations. Given the nature of the work, it was possible for both evaluators to rate a sample of 10 employees at T2. A matched *t*-test indicated that the mean ratings of the two evaluators were equal ($t = 0.21$; $df = 9$; $p = 0.84$). In addition, the agreement between the two evaluators was acceptable, with a Pearson correlation of 0.57.

Results and discussion

The means, standard deviations, and intercorrelations of each of the measures from Study 2 are reported in Table 6.

Affective state and supervisory evaluations

Table 6 shows the correlation of both positive and negative mood with the employee performance ratings. Because mood is considered a transient state, the most relevant test is the relationship of mood to rated performance during the time period in which both variables were measured contemporaneously. For best calibration, the primary emphasis should therefore be placed on the relationships between positive and negative mood and global performance ratings, since each of these scales contained the same temporal referent (i.e. *over the past week*). As shown in Table 6, none of the relationships between mood and rated performance were statistically significant.

Table 6. Means, standard deviations, and Pearson correlations for Study 2*

Variables	<i>N</i>	<i>M</i>	<i>S.D.</i>	1	2	3	4	5	6	7	8	9
1. T1 affective disposition—Berkman	63	3.6	1.3	—	0.77†	0.35‡	−0.27‡	0.47§	−0.44§	0.32‡	0.25‡	0.46
2. T2 affective disposition—Berkman	71	3.4	1.4		—	0.38§	−0.26‡	0.51	−0.64†	0.27‡	0.18	0.43
3. T1 positive mood—PANAS	43	3.1	0.6			—	−0.18	0.62†	−0.28‡	0.09	0.09	−0.01
4. T1 negative mood—PANAS	43	2.0	0.8				—	−0.13	0.40§	−0.18	0.01	−0.03
5. T2 positive disposition—PANAS	53	3.3	0.7					—	−0.43	0.05	−0.05	0.05
6. T2 negative disposition—PANAS	53	1.8	0.7						—	−0.03	−0.01	−0.03
7. T1 global performance ratings	64	3.8	0.7							—	0.69†	0.42
8. T1 composite performance ratings	62	3.5	0.6								—	0.59†
9. T2 composite performance ratings	64	3.4	0.7									—

* One-tailed tests. † $p < 0.0001$; ‡ $p < 0.05$; § $p < 0.01$; || $p < 0.001$.

Pleasantness-based affective disposition and supervisory evaluations

Table 6 also reports the correlations between the Berkman measure of dispositional affect at T1 and T2 and composite performance ratings at T1 and T2. Significant positive relationships were found between this measure of affect at T1 and composite performance ratings at T1 and T2. A significant positive relationship between affect at T2 and composite performance ratings at T2 was also obtained. Similarly, dispositional affect at T1 was significantly related to ratings of global performance at T1.

Table 2 shows the relationship between pleasantness-based affect (T1 and T2) and the composite measure of rated performance at T2. Table 3 shows the correlations between this aggregate measure of affect (T1 and T2) and the four items making up the composite measure of rated performance. Significant positive relationships were obtained between average affect (T1 and T2) and each of the four dimensions at T2.

Though the results in Table 6 involve longitudinal data, they cannot rule out the possibility of various third-variable explanations. For example, Berkman (1971a, pp. 107–110) noted the potential relationships between well-being and such demographic variables as gender, age, job tenure and educational level. Furthermore, while the significant relationships between pleasantness-based affect (T1 and T2) and rated performance (at T2) involved longitudinal data, one cannot rule out the possibility that prior rated performance may have influenced the measure of affect over time.

Multiple regression analysis was performed in order to control for the effects of prior rated performance as well as demographic variables on the affect–supervisory performance evaluations relationship. In this analysis, composite rated performance (T2) was regressed on prior rated performance (T1), gender, age, job tenure, educational level and average pleasantness-based affect (at T1 and T2). Gender and educational level were dummy coded for this analysis ('0' for female, '1' for male; '0' for Bachelor's degree, '1' for post-graduate degree).

Table 7 contains the results of the longitudinal regression analysis. The six independent variables, considered simultaneously, were found to predict subsequent rated performance ($F = 5.866$; $df = 6, 44$; $p = 0.0001$; adjusted $R^2 = 0.37$). The value of the t -statistics for gender ($t = -0.146$; $df = 44$; $p = \text{n.s.}$), age ($t = -0.136$; $df = 44$; $p = \text{n.s.}$), job tenure ($t = -0.343$; $df = 44$; $p = \text{n.s.}$) and educational level ($t = 1.623$; $df = 44$; $p = \text{n.s.}$) did not reach significance. As expected, however, tests for prior rated performance ($t = 3.759$; $df = 44$; $p < 0.001$) and average affect ($t = 1.695$; $df = 44$; $p < 0.05$) revealed a significant contribution to the prediction of subsequent rated performance. These results demonstrate that dispositional affect can significantly predict rated performance over time and when other relevant variables are controlled.

Table 7. Study 2 results of longitudinal multiple regression predicting rated performance at T2

Independent variable	Beta	S.E.	t	p Value (one-tailed)
Rated performance at T1	0.57*	0.15	3.759	<0.001
Average affective disposition (Times 1 & 2)	0.48†	0.28	1.695	<0.05
Age	-0.01	0.07	-0.136	n.s.
Gender	-0.12	0.80	-0.146	n.s.
Tenure	-0.03	0.09	-0.343	n.s.
Education level	1.27	0.78	1.623	n.s.

$R = 0.67$; $R^2 = 0.44$; adjusted $R^2 = 0.37$.

* $p < 0.001$; † $p < 0.05$.

$N = 51$.

Activation-based affective disposition and supervisory evaluations

Table 6 shows the correlation of both positive and negative (trait) affect to the measures of composite rated performance. Neither PANAS scale was significantly correlated with composite rated performance. As shown in Table 8, PANAS scores were also correlated with each of the four items making up the composite measure of rated performance. Out of eight possible correlations, none was statistically significant.

These results clearly demonstrate that neither the trait nor state measure of PANAS was linearly related to supervisory performance evaluations. However, based upon activation theory (Scott, 1966; Champoux, 1980), one might posit a curvilinear relationship between activation and rated performance. Thus, we checked for nonlinear effects using a second-order polynomial model. First, composite rated performance (at T2) was regressed on prior rated performance (at T1) and squared positive affect. Then, composite rated performance (at T2) was regressed on prior rated performance (at T1) and squared negative affect. Neither squared positive affect ($t = 1.429$; $df = 42$; n.s.) nor squared negative affect ($t = -0.003$; $df = 42$; n.s.) were significant predictors in these analyses.

Table 8. Study 2 correlations of affective disposition (PANAS scale) and each rated performance dimension at T2

Positive PANAS at T2 and support at T2:	0.02
Positive PANAS at T2 and goal emphasis at T2:	0.08
Positive PANAS at T2 and team building at T2:	0.08
Positive PANAS at T2 and work facilitation at T2:	-0.02
Negative PANAS at T2 and support at T2:	-0.11
Negative PANAS at T2 and goal emphasis at T2:	-0.08
Negative PANAS at T2 and team building at T2:	-0.02
Negative PANAS at T2 and work facilitation at T2:	0.10

$N = 51$.

Comparative test of affective disposition measures

A multiple regression analysis was performed in order to compare the ability of various formulations of affect in predicting performance. Rated performance at time 2 was regressed on rated performance at T1, pleasantness-based affect, positive activation-based affect, and negative activation-based affect. As shown in Table 9, the only significant predictors of rated performance were prior rated performance ($t = 4.116$; $df = 33$; $p < 0.001$) and pleasantness-based affect ($t = 1.841$; $df = 33$; $p < 0.05$). Activation-based affect did not significantly predict rated performance. These results provide some support for the idea (hypothesis 5) that a pleasantness-based measure of affect is a stronger predictor of rated performance than an activation-based measure of affect.

In general, the results of Study 2 closely paralleled those of the first study. The effects of mood on rated performance were again negligible, even though efforts were taken to temporally calibrate the measurement of mood and rated performance (i.e. *over the past week*). In contrast, the dispositional measure of affect was again a significant predictor of rated performance, both in terms of cross-sectional and longitudinal analyses. Several analyses, however, showed that the difference in predictive power between state and trait affect had more to do with the content of the affect measures than their time frame. Whereas the pleasantness-based Berkman measure showed strong effects in each of the studies, the more activation-based PANAS measure was not predictive of rated performance.

Table 9. Study 2 results of longitudinal multiple regression comparing dispositional affect measures as predictors of rated performance at T2

Independent variable	Beta	S.E.	t	p Value (one-tailed)
Rated performance at T1	0.64*	0.16	3.871	$p < 0.001$
Average affective disposition (Times 1 & 2)	0.85†	0.44	1.909	$p < 0.05$
PANAS positive disposition	-0.03	0.07	-0.382	n.s.
PANAS negative disposition	0.06	0.07	0.783	n.s.

$R = 0.62$; $R^2 = 0.39$; adjusted $R^2 = 0.32$.

* $p < 0.001$; † $p < 0.05$.

$N = 40$.

General Discussion

The hypotheses positing that state affect would covary with performance evaluations were not upheld in either study. Positive mood was more positively associated with rated performance than negative mood, but only one (out of 12) of the correlations from either study reached statistical significance. In Study 1, a measure of rated performance *at this time* was correlated with a *today* measure of state affect. In Study 2, a measure of rated performance *over the past week* was correlated with state affect *over the past week*. In order to create a more precise test of mood's effect on measures of performance, it will be necessary for future research designs to have repeated assessment of mood as well as measures of performance that are reflective of specific changes in behavior. Since few natural organizational settings allow this kind of extensive and objective measurement, such precision will probably require laboratory experimentation.

The results of these studies help to clarify the role of dispositional affect in the happy-productive worker thesis, however. Pleasantness-based dispositional affect was strongly predictive of performance evaluations in both studies, whereas activation-based affect did not predict rated performance. Thus, the notion that the happy worker is also a productive worker may be true, at least when happiness is operationalized by a pleasantness-based measure of dispositional affect.

Although one can argue from these results that pleasantness-based affect is a stronger predictor of rated performance than activation-based affect, such a conclusion could be limited by the nature of performance in various organizational settings. It is possible, for example, that activation-based measures of affect would be more predictive in situations where energy level is a major component of performance. It is also possible that when performance is evaluated from the perspective of the customer rather than the supervisor, aspects of behavior such as being 'attentive' or 'active' could take on more importance. Only by conducting additional research, varying the type of performance and its source of evaluation, will we be able to judge how generalizable the present results are across work roles.

Further measurement issues

Like most organizations, the governmental units in this research relied on supervisory evaluations to measure performance. As a consequence, the meaning of 'performance' could range from concrete observations of work effort to very subjective estimates of an employee's value to the organization. Such ratings of performance may also include halo effects (Cooper, 1981). A broad definition of halo could include any subjective appraisal of employees that is not validated by objective measures of performance. By this definition, one might have to rule out the use of most

performance appraisals in organizations, since seldom is behavior objectively measured, and even when it is, the measures may not tap what actually contributes to organizational effectiveness. A more narrow definition of halo would simply highlight the influence of interpersonal attraction in the appraisal process. Since happy people may be more fun to be around and are more likable, supervisors could have provided more positive evaluations for those who displayed positive affect (Cardy and Dobbins, 1986; Robbins and DeNisi, 1994).

If halo effects were found to be the means by which happy people gain higher performance ratings, this would not negate the basic thesis of the research. The basic fact would remain that those with higher positive affect somehow achieved greater standing or gained more success than others in the organization (something not demonstrated by the earlier research on job satisfaction and performance). What would change would be the implications of these results. Rather than selecting for happy people, organizations would have to guard against those with positive affect gaining more influence and rewards than they deserve.

In their laboratory study of affect–performance relationships, Staw and Barsade (1993) were able to rule out halo effects. They found that those with a more positive disposition performed better on a simulated decision task, even though performance was coded on objective criteria and measures of performance were provided by coders blind to subjects' affective disposition. Because the present field studies used more naturalistic (yet subjective) ratings, we had to rely on both logical inference and some *post-hoc* analyses to deal with the possibility of halo.

As we noted, a halo explanation logically rests on the covariation of affect and ratings of performance. Because our results did not show uniform relationships between all forms of affect and rated performance, a highly specific or idiosyncratic halo effect would be necessary to account for the findings. In addition, some of our analyses seem relatively immune to halo effects. The longitudinal regressions showed the influence of dispositional affect on incremental changes in rated performance over time rather than contemporaneous covariation. Since halo effects should logically be captured in past performance measures (Robbins and DeNisi, 1994), one strength of these longitudinal regressions is the control of a substantial portion of halo.

In an effort to empirically control for halo we also conducted partial correlations between dispositional affect and subsequent rated performance, holding constant the individual's mood measured at the same time as rated performance. The results of these partial correlations were very similar to the zero-order correlations already reported in Table 1. That is, dispositional affect remained a significant predictor of rated performance, after controlling for contemporaneous mood.

As a further check on the likelihood of halo, we also considered how closely affect was related to the various components of rated performance. It could be argued, for example, that support was less task-specific than other components of rated performance such as goal emphasis, making it more susceptible to possible halo effects. In addition, because support measured the extent to which an employee paid attention to the supervisor, it would seem particularly vulnerable to the effects of interpersonal liking. As shown by the data in Table 3, the pattern of results was not especially indicative of a halo effect. We did not find a pattern in which support was the only or by far the strongest relationship with dispositional affect in either study. In contrast, the relationships between affect and rated performance were roughly equal across the four components.

Positive and negative affect

Watson and his associates have strongly argued for the treatment of positive and negative affect as separate conceptual dimensions (Watson and Tellegen, 1985; Watson, Clark and Carey, 1988a). But, as both Watson (1988) and Tellegen (1985) point out, the dimensionality of affect seems to be

highly dependent on the adjective descriptors used in the scales. They noted that many mood terms reflecting happiness ('happy, cheerful') and contentment ('satisfied, pleased, contented') reflect both high positive affect and low negative affect. Conversely, descriptors denoting depression ('sad, blue, gloomy') and loneliness ('alone, lonely') serve as significant markers for both low positive affect and high negative affect. Watson and Tellegen (1985) labeled these descriptors as 'pleasantness terms' and cautioned against using such mixed markers in affect scales. The Berkman (1971a) well-being scale, as we noted, is heavily laden with pleasantness items, integrating both positive and negative descriptors of affect. Thus, one can interpret the findings of this research as indicating that the pleasantness items avoided by Watson's PANAS measure may be precisely the descriptors needed for predicting employee performance.

Although the Berkman scale used to measure dispositional affect was designed as a single bipolar index, we reanalyzed the data to check whether the positive or negative items may have accounted disproportionately for the results. Subscales were created using only the positive or negative items from the Berkman scale and these subscales were then correlated with subsequent ratings of performance. Overall, the positive and negative items showed similar predictive ability. For example, in Study 1, the average of positive affect items (T1 and T2) correlated 0.38 ($p < 0.01$, $n = 52$) with the average of subsequent rated performance (T3 and T4), while the average of negative affect items (T1 and T2) correlated -0.26 ($p < 0.05$, $n = 52$) with the average of subsequent rated performance (T3 and T4). Similarly, in Study 2, the average of positive affect items (T1 and T2) correlated 0.42 ($p < 0.01$, $n = 56$) with rated performance at T2, while the average of negative affect items (T1 and T2) correlated -0.32 ($p < 0.05$, $n = 56$) with rated performance at T2.

Implications for the happy-productive worker

The major goal of these two studies was to find out if various indicators of affect could serve as significant predictors of rated performance. The present data, along with those of Staw and Barsade (1993) and Staw *et al.* (1994), demonstrate that this may indeed be the case.

The most obvious implication of these results is that organizational researchers may have prematurely put to rest the notion of the happy-productive worker. Although this popular thesis has not been validated with job satisfaction research, there may yet be some truth to the idea when pleasantness-based dispositional affect is considered. In fact, it is even possible that managers and the lay public have persisted in their belief in the happy-productive worker precisely because of covariation between dispositional affect and measures of performance. Ironically, while managers may have been observing employees who were both productive and dispositionally positive, researchers have been looking in vain for links between job satisfaction and various measures of performance. Thus, a shift in research direction is called for. Future research should now examine whether a syndrome exists in which dispositional affect and performance are associated, much as the personality traits labeled 'conscientiousness' have been found to covary with performance (Barrick and Mount, 1991).

Certain cautions should of course be taken regarding the implications of our findings. Without further research involving multiple measures of performance, it will be difficult to know exactly *why* positive people are more successful in organizations. Staw *et al.* (1994) eschewed this issue by noting that, even if supervisory evaluations include halo or other forms of bias, they are predictive of achievement or success *from the employee's point of view*. While such an argument is valid, we also believe that it is important to understand the reasons underlying a person's success. Now that we have verified the existence of an affect-performance relationship, it is time to

investigate the mechanisms responsible for such a linkage. Future research therefore needs to examine whether the success of positive people is due to particular interpersonal behaviors (e.g. helping, being encouraging, use of ingratiation techniques), aspects of their task motivation (e.g. effort, persistence, higher goals), or simple illusion (i.e. halo) on the part of those allocating rewards in organizations. Sorting out these various paths to 'success' will take a concerted research effort, one that will no doubt involve measures of the work process as well as products attained by people differing in affective disposition.

The person versus situation debate revisited

In closing, we should note that the results of these studies also have implications for the debate on dispositional versus situational influence now central to organizational behavior (Arvey, Bouchard, Segal and Abraham, 1989; Davis-Blake and Pfeffer, 1989; George, 1992; Gerhart, 1987; Staw and Ross, 1985; Wright *et al.*, 1993). The effect of dispositional affect on subsequent ratings of performance is supportive of the argument that individual characteristics can be a potent influence on behavior and outcomes over time. It is not yet clear, however, whether this influence is primarily affective or cognitive in nature. We do not know for certain whether continuities in emotional temperament, such as positive and negative affect, are primary forces in a person's behavior over time. Or, whether cognitive tendencies, such as those tapped by Seligman's (1991) construct of optimism, are more basic to the kinds of actions and decisions people make in organizations. This uncertainty is closely tied to a dispute in psychology over the primacy of affect versus cognition. Some would argue (e.g. Zajonc, 1984), that once affect is determined, cognitive tendencies are necessarily yoked along. Others would contend (e.g. Lazarus, 1984) that cognitive tendencies, such as optimism, are precursors of affective reactions toward various stimuli in social and work settings. A third position might suggest that positive affect and optimism are simply two sides of a larger personality trait, one that is so intertwined that it makes little sense to separate its affective and cognitive components. How this dilemma gets resolved will no doubt shape future research on both dispositions and emotion in organizational life.

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